

The irrationality exponent of positive quadratic logarithms is at most 4

Ryan Matthew Casper

October 8, 2026

Abstract

For every positive real algebraic number of degree two other than one, we prove that its logarithm has irrationality exponent at most four. The argument extends weighted logarithmic interpolation and determinant estimates to the original quadratic field. Integer clearing, the field norm, and a separate bound at the other embedding accommodate nonintegral inputs and negative conjugates. The denominator threshold is not made effective. A Lean formalization accompanies the proof.

1. Target, conventions, and algebraic setup

Let $a \in \mathbb{R}$ satisfy $a > 0$, $a \neq 1$, and $[\mathbb{Q}(a) : \mathbb{Q}] = 2$. Put $\xi = \log a$. The target is

$$\forall \nu > 4 \exists Q \in \mathbb{N} \forall p \in \mathbb{Z} \forall q \in \mathbb{N}, \quad q \geq Q \implies \left| \xi - \frac{p}{q} \right| \geq q^{-\nu}. \quad (1)$$

We may and will require $Q \geq 2$. The threshold may depend on a, ν ; no effectivity or uniformity in a is asserted. Fractions need not be reduced. With the usual irrationality exponent defined using errors $|\xi - p/q| < q^{-\nu}$, (1) implies $\mu(\xi) \leq 4$.

Write $L = \mathbb{Q}(a) \subset \mathbb{R}$. Its other embedding is real: if the monic minimal polynomial is $X^2 + tX + u$, its other root is $a' = -t - a \in \mathbb{R}$. Irreducibility makes $a' \neq a$ and gives the nontrivial field automorphism $\tau : L \rightarrow L$, $\tau(a) = a'$. No positivity assumption is made on a' . Choose an integer $d \geq 1$ for which da is an algebraic integer. Such a choice follows, for example, by taking the absolute value of a nonzero leading coefficient of an integer polynomial for a .

For a nonzero algebraic integer $z \in L$,

$$z\tau(z) = N_{L/\mathbb{Q}}(z) \in \mathbb{Z} \setminus \{0\}, \quad |z| |\tau(z)| \geq 1. \quad (2)$$

These are standard quadratic-field facts. They are not consequences of an absolute-value lower bound at either embedding separately.

We argue by contradiction for a fixed $\nu > 4$. If (1) fails, there are approximants with arbitrarily large denominators and

$$|\xi - p/q| \leq q^{-\nu}. \quad (3)$$

Set $R = |\xi| + 1$ and $\rho = 100 \max(1, |\xi|)$. Every chosen approximant with $q \geq 2$ has $|p/q| \leq R$. Since $\xi \neq 0$, sufficiently large denominators also ensure $p \neq 0$, if that convenient extra condition is required by a wrapper.

2. Exact inherited inputs

All source paths in this section are relative to the pinned rational-logarithm dependency project. The namespace prefix `0AI` is suppressed below where unambiguous. These statements have no hypothesis that the multiplicative centres or the target logarithm are rational.

2.1 Interpolation

Fix integers $m, K \geq 1$, positive rational degree weights $W = (w_0, w_1, \dots, w_m)$, positive rational jet weights $V = (v_0, w_1/\theta, \dots, w_m/\theta)$, $0 < \theta < 1$, and rational $\sigma > 0$. Require

$$K(1 + 3\sigma)^{m+1} \frac{\prod W_i}{\prod V_i} < 1, \quad (1 + \sigma)\theta < 1, \quad (4)$$

and the separated-product inequalities in `Logarithm.GeometryData.curve_separated_weight_products`: for equal-size subsets $I, J \subseteq \{0, \dots, m\}$, if their largest differing index is positive and belongs to I ,

$$\mathcal{C}(m, \sigma) \prod_{j \in J} V_j < \prod_{i \in I} W_i, \quad \mathcal{C}(m, \sigma) = \text{comparisonConstant}(m, \sigma). \quad (5)$$

We use the actual source constant; only its finiteness and independence from the centres matter here.

For arbitrary distinct nonzero $y_j \in \mathbb{C}$ and arbitrary $c_{ji} \in \mathbb{C}$, the coefficient packets

$$[t^s u^\beta] P(y_j(1+t), c_{j1} + u_1 + \log(1+t), \dots, c_{jm} + u_m + \log(1+t)), \quad v_0 s + w\beta/\theta < H,$$

can all be prescribed by polynomials of W -degree at most H , for all sufficiently large $H = n\mathcal{R}$. The positive rational scale $\mathcal{R}$ depends only on the weights; the eventual threshold may depend on the fixed centres.

Declaration: `Logarithm.eventually_weighted_logarithmic_interpolation` in `Logarithm/Interpolation.lean`, applied to the concrete `GeometryData`. Its conclusion is the surjectivity of `GeometricInterpolation.packetMap`; `GeometricInterpolation.WeightedPolynomials` supplies the degree bound. Replacing each logarithm by a truncation G_i is allowed when $T_i v_0 \geq w_i/\theta$, by `eventually_weighted_truncated_interpolation`. The conversion to the literal matrix is `Logarithm.ScaledMatrix.truncatedLogMatrix_surjective_of_formalLog_packets` in `ScaledMatrix.lean`.

The substantial geometric proof—curve inequality, blowup ampleness, eventual jet restriction, and polynomial section degree bound—is inherited through these declarations. This manuscript does not replace it by a dimension count.

2.2 Generic analytic determinant ingredients

The following declarations are used with $\omega = \xi \in \mathbb{C}$, rational additive centres $r_i = p_i/q_i$, and the entry factor a^{jh} .

- Finite row expansion with the centre factor retained: `Logarithm.ScaledTranslation.det_matrix_translation`, `Logarithm/ScaledTranslation.lean`
- Error $j(r_i - \xi)$ in exponential weight form: `Logarithm.ScaledApproximationErrors.actual_error_exp`, `Logarithm/ScaledApproximationErrors.lean`

- Translation scalar estimate: `PiExponent.MatrixTranslationBounds.norm_rowScalar_exp_weight_difference`, `OAI/NumberTheory/PiExponent/Approximation/MatrixTranslationBounds.lean`
- Generic collision estimate: `Logarithm.ScaledCollision.formal_scaled_collision_bound`, `Logarithm/ScaledCollision.lean`
- Conversion of formal period coefficients to analytic row tests: `Logarithm.ScaledPeriodAnalytic.scaled_periodMonomial_coeff_eq_rowTest`, `Logarithm/ScaledPeriodAnalytic.lean`
- Two alternatives and finite summation: `PiExponent.DeterminantAnalyticBound.translated_summand_bound` and `log_norm_sum_le`, `OAI/NumberTheory/PiExponent/Analysis/DeterminantAnalyticBound.lean`
- Row and low-index lattice asymptotics: `PiExponent.MatrixCounting.tendsto_rowCount_normalized`, `tendsto_lowIndexCount_normalized`, `tendsto_collisionRatio`, `OAI/NumberTheory/PiExponent/Approximation/MatrixCounting.lean`

The rational wrappers `LiteralAnalytic.actual_minor_analytic_bound` and `GlobalMatrixInterpolation.cofinal_actualMatrix` **cannot be applied directly** to a . Sections 3 and 5 below reassemble their generic ingredients and discharge the hypotheses. No new analytic theorem is being silently assumed.

3. Parameters are chosen before approximants

Let $\Lambda = 4 + \log 4$, $c = \log 2/4$. Choose rational θ, A, B, C and real η satisfying

$$0 < \theta < A < B < 1, \quad C > 1, \quad CB < 1, \quad B < C\theta < 1, \quad 0 < \eta < 1, \\ g := \nu(A(1 - \eta) - \theta) - 2(1 - \theta) > 0. \quad (6)$$

This follows directly from `PiExponent.exists_parameters` (`nu/2`) since $\nu/2 > 2$: multiply its final gap by two. For an explicit elementary construction, choose rational $b \in (1/2, 1 - 2/\nu)$, put $\theta = 1 - s$, $A = 1 - bs$ with sufficiently small rational $s > 0$, use $A^2 < \theta$ to choose $C \in (A/\theta, 1/A)$, and then choose $A < B < \min(C\theta, 1/C)$. A sufficiently small positive η preserves the strict gap.

Fix $0 < \varepsilon < \min(g, 1)$. Choose $F > 2/\theta$ such that $\nu/F < \varepsilon/3$. Set

$$K = \lfloor C^m \rfloor, \quad w_0 = B^{-m}, \quad v_0 = 2K\theta^m w_0, \quad L' = \log \max(1, |a'|), \quad \delta = \log d. \quad (7)$$

All w_0, v_0 are positive rational numbers, and

$$K(w_0/v_0)\theta^m = 1/2, \quad K\theta^m < 1.$$

As $m \rightarrow \infty$, $K/w_0 \rightarrow 0$, $1/w_0 \rightarrow 0$, $m/v_0 \rightarrow 0$, while

$$c_\infty := \frac{c\eta^2 K\theta^m}{(m+1)v_0 A^m} = \frac{c\eta^2 (B/A)^m}{2(m+1)} \rightarrow \infty. \quad (8)$$

The floor in K causes no problem: $\lfloor C^m \rfloor / C^m \rightarrow 1$. These are also the identities underlying `exists_dimension_margin` in the inherited parameter machinery.

Choose one finite $m \geq 1$ for which $c_\infty > 3$ and

$$\frac{2\Lambda Fm + 3\log 2}{v_0} + \frac{(\rho + 2\delta + L')K + \log(3/2)}{w_0} < \varepsilon/3. \quad (9)$$

Freeze m, K, w_0, v_0 . Choose rational $\sigma > 0$ sufficiently small that (4) holds; one may also impose the inherited optional condition $(1 + 3\sigma)^m K \theta^m < 1$.

Only now choose m approximants from (3), successively, with $q_i \geq 2$ and integer weights $w_i = \lceil \log q_i \rceil$. Arbitrarily large denominators allow the w_i to be as large as desired. They can satisfy (5): cancel common factors in each product comparison and solve for its largest differing weight, whose index belongs to the degree-weight side; all remaining factors have smaller indices and are already fixed. There are finitely many such comparisons at each step. These thresholds depend on previously chosen weights, not on the centre locations.

Also make $w_* := \min_i w_i$ large enough that

$$2\Lambda \sum_i \frac{1}{w_i} + \frac{2\theta + \log 4 + \log(2K) + \nu + \log(2\rho K) + \log(KR + 2)}{w_*} < \varepsilon/3. \quad (10)$$

This is possible because the finite numerator in (10) was fixed before choosing the approximants, and $\sum_i 1/w_i \leq m/w_*$.

Finally set

$$T_i = \lceil Fw_i/v_0 \rceil, \quad G_i(t) = \sum_{1 \leq k < T_i} \frac{(-1)^{k+1} t^k}{k}.$$

Thus $T_i v_0 \geq Fw_i > w_i/\theta$. All parameters, approximants, truncations, and centres are now frozen. The only remaining limit is $H \rightarrow \infty$.

4. One selected algebraic minor and its arithmetic bound

Take $y_j = a^j$, $c_{ji} = jr_i$, $0 \leq j < K$. The y_j are distinct and nonzero because $a > 0$, $a \neq 1$. The generic interpolation input therefore gives full row rank at every sufficiently large admissible height $H = n\mathcal{R}$.

Rows are (j, s, β) with $v_0 s + w\beta/\theta < H$; columns are (h, γ) with $w_0 h + w\gamma \leq H$. All indices are nonnegative integers. The entry is

$$E_{(j,s,\beta),(h,\gamma)} = [t^s u^\beta] a^{jh} (1+t)^h \prod_i (jr_i + G_i(t) + u_i)^{\gamma_i} \in L. \quad (11)$$

Choose a full-row square minor $\Delta_H \neq 0$, with M_H rows. This selection is fixed in every estimate at that height. Its conjugate is the same selection with a replaced by a' ; it is nonzero because τ is injective.

Define

$$b_H = \frac{\sum_{\text{rows}} w\beta}{M_H H}, \quad 0 \leq b_H \leq \theta.$$

There is at least one row for $H > 0$, so $M_H H > 0$. Let $L_i = \text{lcm}(1, \dots, T_i)$ and

$$D_H = \prod_i L_i^{\lfloor H/w_i \rfloor}.$$

Multiply each entry by D_H , its row factor $\prod_i q_i^{-\beta_i}$, and its column factor $d^{Kh} \prod_i q_i^{\gamma_i}$. If $\beta \not\preceq \gamma$, the entry is zero. Otherwise it becomes

$$D_H (da)^{jh} d^{(K-j)h} \binom{\gamma}{\beta} [t^s] (1+t)^h \prod_i (jp_i + q_i G_i(t))^{\gamma_i - \beta_i}. \quad (12)$$

Every coefficient of $L_i G_i$ is an integer. Since $\gamma_i - \beta_i \leq \gamma_i \leq \lfloor H/w_i \rfloor$, the factor D_H clears all polynomial coefficient denominators in (12). Thus (12) is an algebraic integer in L , not merely in $L(i)$. The determinant of the scaled matrix is

$$Z_H = C_H \Delta_H, \quad C_H = D_H^{M_H} \prod_{\text{selected columns}} \left(d^{Kh} \prod_i q_i^{\gamma_i} \right) \prod_{\text{rows}} \prod_i q_i^{-\beta_i} > 0. \quad (13)$$

The positive rational number C_H need not be integral. The scaled entries and their determinant are integral, which is what (2) needs. Consequently

$$1 \leq C_H^2 |\Delta_H| |\tau \Delta_H|. \quad (14)$$

For completeness, the clearing cost is

$$\frac{\log C_H}{M_H H} \leq 1 - b_H + E_{\text{ar}}, \quad E_{\text{ar}} = \frac{\Lambda F m}{v_0} + \Lambda \sum_i \frac{1}{w_i} + \frac{\theta}{w_*} + \frac{K\delta}{w_0}. \quad (15)$$

Indeed $\log L_i \leq \Lambda T_i$, $T_i \leq F w_i / v_0 + 1$, and $\lfloor H/w_i \rfloor \leq H/w_i$, giving the first two terms. A selected column satisfies $\sum_i \gamma_i \log q_i \leq \sum_i \gamma_i w_i \leq H$, and $h \leq H/w_0$, giving $1 + K\delta/w_0$. For each row, $\log q_i \geq w_i - 1$ and $\sum_i \beta_i \leq w\beta/w_* \leq \theta H/w_*$. Thus its subtracted cost is at least $w\beta - \theta H/w_*$, proving (15). This exact row subtraction is preserved twice in (14).

The lcm estimate is `PiExponent.Arithmetic.log_lcmUpto_le`; the column and row inequalities are `column_log_cost_le` and `row_log_cost_lower` in `Approximation/Arithmetic.lean`, or their matrix wrappers. The new ordinary-integer version of (12) is the isolated module `AlgebraicLog/IntegerClearing.lean`; the inherited Gaussian clearing endpoint is deliberately not used to infer integrality in L .

5. Distinguished analytic bound, with all substitutions

Put

$$f_{b,h,\gamma}(z) = [u^b] e^{hz} \prod_i (z + u_i)^{\gamma_i} = \binom{\gamma}{b} e^{hz} z^{|\gamma| - |b|}$$

when $b \leq \gamma$, and zero otherwise. These are entire functions. At row centre j , write $z_j(t) = j\xi + \log(1+t)$. Since $e^\xi = a$, $e^{z_j(t)} = a^j(1+t)$ for $|t| < 1$. The additive coordinates in (11) are

$$z_j(t) + u_i + e_{ji} + \tau_i(t), \quad e_{ji} = j(r_i - \xi), \quad \tau_i = G_i - \log(1+t).$$

The finite row expansion in the source list above therefore applies literally. A row choice consists of a transverse multi-index b with $wb \leq H$, a distribution of error/tail factors, and a coefficient order $k \leq s$. The scalar is independent of the selected column. Nonzero scalar terms have $b \geq \beta$.

From (3) and $w_i - 1 \leq \log q_i$, $|e_{ji}| \leq K e^\nu e^{-\nu w_i}$. The tails begin in order T_i . Apply `norm_rowScalar_exp_weight_difference` with its error constant $\log(2K) + \nu$, weights w_i , scale F , lower weight w_* , and $k \leq s < H/v_0$. Its hypotheses hold because $T_i \geq 1$, $F w_i \leq v_0 T_i$, and the approximation bound just proved. It gives the scalar bound

$$\exp\{-\nu(wb - w\beta) + H E_{\text{tr}}\}, \quad E_{\text{tr}} = \frac{\nu}{F} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(2K) + \nu}{w_*}. \quad (16)$$

Apply `formal_scaled_collision_bound` with row label j , row order $\ell = s - k$, transverse grouping b , column exponents h, γ , $\omega = \xi$, and ρ chosen above. All its hypotheses follow from

$0 \leq j < K$, the column degree budget, $\sum \ell \leq M_H H / v_0$, $\rho \geq 100$, and $\rho \geq 100|\xi|$. Its centre factor $e^{j\xi^h}$ equals exactly a^{jh} . For n_b rows in each group, the resulting determinant is at most

$$\exp \left\{ -c \sum_b n_b^2 + M_H H (E_{\text{hol}} + o(1)) \right\}, \quad E_{\text{hol}} = \frac{\rho K}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2\rho K)}{w_*}. \quad (17)$$

The source writes the last logarithm as $\log(200K) + \log(\rho/100)$, which is equal. The $o(1)$ is `Collision.collisionRemainder M_H H`; its convergence follows from the positive row-count asymptotic below and `tendsto_collisionRemainder_of_normalized_pow`.

Let $N_A(H) = \#\{b : wb \leq AH\}$ and

$$c_H = \frac{c\eta^2 M_H}{HN_A(H)}.$$

If at least ηM_H rows use low indices, Cauchy–Schwarz gives $\sum n_b^2 \geq \eta^2 M_H^2 / N_A(H)$. Otherwise the total index weight is at least $A(1 - \eta)M_H H$. Also $\sum wb \geq \sum w\beta = M_H H b_H$ for every nonzero scalar product. Applying `translated_summand_bound` gives exactly these two alternatives.

There are at most $Q_H = (\lfloor H \rfloor + 1)^{2m} (\lfloor H/v_0 \rfloor + 1)$ choices per row, as in `MatrixTranslation.row_term_count`; $w_i \geq 1$ and $s \leq H/v_0$ discharge its hypotheses. Sum the at most $Q_H^{M_H}$ terms using `log_norm_sum_1e`. Since $\log Q_H/H \rightarrow 0$, we obtain, uniformly in the selected columns,

$$\frac{\log |\Delta_H|}{M_H H} \leq E_{\text{an}} + o(1) + \max\{-c_H, -\nu(A(1 - \eta) - b_H)\}, \quad E_{\text{an}} = E_{\text{tr}} + E_{\text{hol}}. \quad (18)$$

The rational positive weights and $q_i \geq 2$ allow the lattice-counting declarations in the table to be applied, giving

$$M_H \sim \frac{K\theta^m H^{m+1}}{(m+1)!v_0 \prod_i w_i}, \quad N_A(H) \sim \frac{A^m H^m}{m! \prod_i w_i}, \quad c_H \longrightarrow c_\infty. \quad (19)$$

Thus every normalized error in (18) tends to zero after all other data have been fixed.

6. The other embedding: a new coarse bound

Apply the same coefficient formula (11) with a' in place of a . On $|t| = 1/2$, $|G_i(t)| \leq \sum_{k \geq 1} 2^{-k}/k = \log 2$. On the additional circles $|u_i| = 1$,

$$|jr_i + G_i(t) + u_i| \leq KR + \log 2 + 1 < KR + 2.$$

Repeated one-variable Cauchy coefficient bounds therefore give

$$|\tau E_{(j,s,\beta),(h,\gamma)}| \leq 2^s \max(1, |a'|)^{Kh} (3/2)^h (KR + 2)^{|\gamma|}.$$

The u -circle radii equal one, so no additional β -dependent cost appears. This estimate already includes all binomial coefficients. Use $s < H/v_0$, $h \leq H/w_0$, and $|\gamma| \leq H/w_*$ to bound every entry by e^{HU} , where

$$U = \frac{\log 2}{v_0} + \frac{KL' + \log(3/2)}{w_0} + \frac{\log(KR + 2)}{w_*}. \quad (20)$$

The permutation formula for the determinant then gives

$$\frac{\log |\tau \Delta_H|}{M_H H} \leq U + \frac{\log(M_H!)}{M_H H} = U + o(1). \quad (21)$$

Indeed $M_H = O(H^{m+1})$, so $\log(M_H!)/(M_H H) \leq \log M_H/H \rightarrow 0$. This proof is uniform in the selected minor. It uses neither $\log a'$, an approximation to it, nor any relation between a' and a^{-1} .

Combining (14), (15), and (21) proves the new arithmetic lower bound

$$\frac{\log |\Delta_H|}{M_H H} \geq -2(1 - b_H) - 2E_{\text{ar}} - U - o(1). \quad (22)$$

6.1 Conservative coefficient bound used by the Lean development

The module `ConjugateBound.lean` instead uses the convenient one-variable estimate $|G_i(t)| \leq 2$, obtained from the inherited bounds for $\log(1+t)$ and its tail. Bounding $\binom{\gamma}{\beta} \leq 2^{|\gamma|}$ then replaces $KR + 2$ in (20) by $2(KR + 2)$. This proves a slightly weaker bound $\tilde{U} = U + \log 2/w_*$. For exact agreement with that implementation, replace $\log(KR + 2)$ in (10) and the total error below by $\log(2(KR + 2))$, and use $\tilde{U}$ in (21)–(22). All parameter choices and the final argument are unchanged: the additional fixed numerator is chosen before w_* . This variant avoids a multivariate Cauchy formalization and still gives exponent four.

6.2 A fixed bound for every embedding

The formal parameter construction uses the slightly larger constant

$$\kappa = \sum_{\sigma: L \hookrightarrow \mathbb{C}} \log \max(1, |\sigma(a)|)$$

in place of L' . Every summand is nonnegative, so this constant bounds the required other-embedding cost. It is fixed before the dimension is chosen, and hence the same dimension limit absorbs it. The norm argument runs over all complex embeddings of the actual field $\mathbb{Q}(a)$. Since its degree is two, removing the distinguished embedding leaves exactly one term. This avoids choosing a named conjugate or assuming its sign, while preserving the exact coefficient two on the rational clearing factor.

7. Contradiction and endpoint

Let $E = 2E_{\text{ar}} + U + E_{\text{an}}$. Expanding (15)–(17) and (20) yields exactly

$$\begin{aligned} E = & \frac{\nu}{F} + \frac{2\Lambda Fm + 3\log 2}{v_0} + \frac{(\rho + 2\delta + L')K + \log(3/2)}{w_0} \\ & + 2\Lambda \sum_i \frac{1}{w_i} + \frac{2\theta + \log 4 + \log(2K) + \nu + \log(2\rho K) + \log(KR + 2)}{w_*}. \end{aligned}$$

By our choices, $E < \varepsilon < \min(g, 1)$ and $c_\infty > 3 > 2 + E$. Comparing (18) and (22) forces at least one of

$$c_H \leq 2(1 - b_H) + E + o(1), \quad (23)$$

$$\nu(A(1 - \eta) - b_H) \leq 2(1 - b_H) + E + o(1). \quad (24)$$

The first is impossible for all sufficiently large H , since $b_H \geq 0$ and $c_H \rightarrow c_\infty > 2 + E$. For the second,

$$\nu(A(1 - \eta) - b_H) - 2(1 - b_H) = g + (\nu - 2)(\theta - b_H) \geq g > E,$$

so it too is impossible for all sufficiently large H . The $o(1)$ terms are uniform in the selected columns and b_H ; take a fixed margin smaller than both strict gaps. Interpolation supplies arbitrarily large admissible H , completing the contradiction.

Thus (1) holds. Moreover ξ is irrational: if $\xi = u/v$, the unreduced fractions $ku/(kv)$ violate (1) for arbitrarily large denominators. Therefore the usual exponent is defined on an irrational real and is at most four. Its defining exponent set is nonempty by Dirichlet and bounded above by four; if using a real supremum in Lean, those facts must be supplied explicitly. We have not proved the endpoint estimate with exponent exactly four and a fixed constant.

8. Nonvacuity and checked declarations

The first example is

$$a_1 = (1 + \sqrt{2})/2, \quad a'_1 = (1 - \sqrt{2})/2, \quad N(a_1) = -1/4.$$

It is positive, not one, quadratic, and nonintegral; $d = 2$ gives $2a_1 = 1 + \sqrt{2}$, an algebraic integer. Here $|a'_1| < 1$, so $L' = 0$. A stronger arithmetic-cost test is

$$a_2 = (1 + 3\sqrt{2})/2, \quad a'_2 = (1 - 3\sqrt{2})/2, \quad N(a_2) = -17/4.$$

Again $d = 2$ works, while $a'_2 < -1$ and $L' = \log((3\sqrt{2} - 1)/2) > 0$. Neither test has norm one; the second exercises the nonzero conjugate-growth cost. Their integer polynomials are $4X^2 - 4X - 1$ and $4X^2 - 4X - 17$, respectively. Irreducibility follows from irrationality of $\sqrt{2}$.

The accompanying source directory is `lean/AlgebraicLog/`. All names in the following list have prefix `OAI.AlgebraicLog`.

- Ordinary integer coefficient clearing: `IntegerClearing.entry_truncatedLog_cleared_int`
- One integral minor and the same determinant at every embedding: `MatrixArithmetic.exists_integral_minor_det_lift`
- Actual quadratic norm inequality: `QuadraticNorm.quadratic_literal_minor_log_bound`
- Normalized arithmetic bound with the row saving doubled: `SelectedArithmetic.normalized_selected_lower`
- Actual conjugate determinant bound and vanishing remainder: `ConjugateAggregate.complexMinor_log_bound, tendsto_conjugateRemainder`
- Full numerical parameters from actual- ν approximants: `exists_admissible_parameters`
- Distinguished analytic bound: `LiteralAnalytic.actual_minor_analytic_bound`
- Cofinal interpolation of the actual matrix: `GlobalMatrixInterpolation.cofinal_actualMatrix`
- Contradiction for actual algebraic data: `DeterminantContradiction.no_fixed_data`
- Actual rational adjunction and integer clearing: `FieldRealization.realField_finrank, exists_positive_integral_multiple`
- Literal quantified endpoint (1): `quadratic_log_explicit_bound`
- Conventional irrationality-exponent consequence: `quadratic_log_irrationalityExponent_le_four`
- Both nontrivial test inputs: `Examples.first_sample_bound, Examples.second_sample_bound`

The final endpoint uses the standard hypotheses `IsAlgebraic Q alpha` and `(minpoly Q alpha).natDegree = 2`, not a custom data record. The parameter record is constructed inside

the proof from a hypothetical failure of (1). The real-supremum consequence proves irrationality from the unreduced lower bound, uses Dirichlet to supply a nonempty exponent set, and proves that every member is at most four. The two examples' algebraicity and degree are also proved, rather than assumed.

The accompanying audit prints the final statements and axiom lists and checks the explicit target proposition. The verification record describes native builds, exported-proof checks, and automated reviews, together with their dependency trust boundaries. See [verification and reproduction](#).

No invariant has been transported through multiplication by an irrational algebraic number. No norm-one theorem, Galois averaging of approximation estimates, or auxiliary field extension is used. Mathematical novelty and external human verification remain separate questions from this development.

AI use

The author initiated and directed an investigation of extending OpenAI's irrationality-exponent method to logarithms of positive quadratic algebraic numbers. OpenAI coding agents contributed substantially to developing the candidate argument, producing the Lean formalization, and conducting automated adversarial reviews. The construction builds on OpenAI's published manuscript and released Lean library, the preceding rational-logarithm extension, and Mathlib. Automated checking and model review are distinct from independent human mathematical review.

References and attribution

1. OpenAI, *The irrationality exponent of π is 2*, manuscript dated September 24, 2026, and [released \$\pi\$ library](#). The weighted interpolation and analytic determinant framework comes from this development.
2. Ryan Matthew Casper, *The irrationality exponent of positive rational logarithms is 2*, October 7, 2026, [pinned manuscript and formalization](#). This supplies the inherited moving-centre interpolation and logarithmic matrix interfaces used here.
3. The mathlib Community, [Mathlib](#), pinned revision used by the formalization.

The proposed contribution is the general quadratic upper bound and its formal extension. Irrationality of the logarithm of a nontrivial positive algebraic number is classical. This paper makes no exhaustive novelty or priority claim.